SELLAMUTHU R

krssellamuthu2007@gmail.com (+91) 7305537699 SellamuthuRamu sellamuthu2007

EDUCATION

Chennai Institute of Technology - Chennai 2024–2028
Bachelor of Engineering Current **CGPA: 8.78/10.0**
Coursework : Computer Architecture, Computer Networks, Operating Systems, Software Engineering, DBMS

EXPERIENCE

BETSHOOW - Application Development Intern NOV 2025 – DEC 2025
India

- Completed Frontend Development Internship working with React.js and Tailwind CSS.
- Improved UI performance and strengthened API integration for the radiology management system.
- Developed reusable React components and enhanced user experience through responsive UI design.

PROJECTS

COURT CASE MONITORING SYSTEM (CCMS) – Legal Tech & Automation Platform – React, FastAPI, PostgresSQL.

- Built a responsive dashboard supporting automated status tracking, notifications, and personalized action feeds.
- Implemented secure role-based auth, user profiles, and cloud storage for legal document management.
- Integrated AI-powered parsing APIs and state management for smooth data processing and device optimization.

STAFF ATTENDANCE – Staff Duty & Attendance Monitoring System - React, JavaScript, Supabase, *Git*.

- Developed a staff attendance platform for tracking check-ins, duty hours, proxy detection, and maintaining records.
- Implemented secure authentication, role-based access control, and dashboards to monitor staff activities.
- Integrated APIs and efficient state management to ensure smooth interactions and optimized performance.

AGENTIC AI FINANCIAL ANALYST – Python, Phidata, Groq, GPT-OSS 120B, YFinance, DuckDuckGo, FastAPI

- Built a multi-agent financial analysis system using **Phidata** and **Groq GPT-OSS 120B**, combining specialized web-search and financial-analysis agents for automated stock research.
- Integrated **YFinance** for stock prices, fundamentals, analyst recommendations, and company newss, with **DuckDuckGo** for live web research and source-backed financial insights.

EMPLOYEE HANDBOOK RAG – Python, RAG, PyPDF, Sentence Transformers, BAAI/bge-m3, NumPy, ChromaDB, Groq.

- Developed a **Retrieval-Augmented Generation (RAG)** system that extracts employee handbook content using **PyPDF**, applies overlapping chunking, and generates semantic embeddings using **BAAI/bge-m3** with Sentence Transformers.
- Implemented persistent **ChromaDB** vector storage and top-3 similarity retrieval, followed by **Groq** generation to produce context-grounded answers from retrieved handbook content.

SKILLS

- **Programming Languages:** C, C++, JavaScript, SQL, Python.
- **Artificial Intelligence:** Pandas, NumPy, Matplotlib, LangChain, LangGraph, RAG, Seaborn, OpenAI API, Hugging Face.
- **Frontend and Design:** HTML, CSS, React.js, Next.js, Tailwind CSS, Django, Canva.
- **Backend and Databases:** Node.js, Express, MySQL, Supabase, MongoDB, SQLite, Firebase.
- **CS Fundamentals:** Data Structures, Algorithms, Computer Architecture, Operating Systems, DBMS, OOPs.
- **Soft Skills and Tools:** Leadership, Team Management, Communication, Git, GitHub, VSCode, IntelliJ, Postman.

COMPETITIVE PROGRAMMING

- LeetCode: Max rating - 1804 | Solved problems: 1000+ | Top 9% globally | Badges: 6 | Highest Rank: 4745.
- CodeChef: Max rating - 1321 | Position : 1 Star | Problems: 530+ | Global rank : 3811 in Starters 243 (Rated).
- Codeforces: Max rating - 866 | Position: Newbie | Solved problems: 110+ | Highest rank : 7173 in round 1085.

ACHIEVEMENTS

- **Tesco Retail Jam :** Built a GenAI-powered retail media creative tool, Finalist - 3K Credits, **Tesco Retail Media.**
- **CSIC Hackathon :** Built an e-KYC verification system using cryptographic keys (Top 50), **DSCI - Govt of India.**
- **IDEATHON :** Built an AI-powered price prediction app to support regional languages, **IIITDM-Kancheepuram.**

CERTIFICATION

- MongoDB - MongoDB (Basics)
- NPTEL - Introduction to IoT
- AWS - AWS Cloud Practitioner Essentials
- Cisco - Introduction to Networks
- Cisco - Introduction to Cybersecurity
- IBM - AI Fundamentals with IBM

VOLUNTEERING

- Assisted in organizing and coordinating the Hackerz Code-A-Thon hackathon, supporting event operations and ensuring smooth participation and engagement.